

Exercise12

15.25 / 18 + 2 Bonus

CPS I — Exercise Sheet 12

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Author: Zhanyu Dong, Xin Pu

Exercise 1 :Done 2/2

Exercise 2: Fairness 9.25/12

F1 : For (a) to (f) all false. explanations? it says on the task to "justify your answer" ~ 0.25 * 6

F2 ^{what is it?}
a: Yes, it is not enabled.

b: Yes, δ is enabled and executed infinitely often

c: Yes, we have execution. $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\delta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta})^\omega$ It satisfies weak fairness for δ ✓

d: No, δ is infinite enabled since s_1 , but only execute only once. ✓

e: Yes, the same in (c). We have execution. $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\delta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta})^\omega$ It satisfies weak fairness for δ ✓
extension to fair path would be: $s_0 s_2 s_1 (s_2 s_1)^\omega$

f: Yes, the same in (c). We have execution. $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\delta} (s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta})^\omega$ It satisfies weak fairness for δ ✓
~ 0.25

Stated in words, a fairness assumption is **realizable** in a transition system TS whenever in any reachable state at least one fair execution is possible. This entails that every initial finite execution fragment of TS can be completed to a fair execution. Note that there is no requirement on the unreachable states.

F3:

a: Yes, not enabled. ✓

b: No, infinite enabled but not executed. ✓

c: Yes, we have execution $s_0 \xrightarrow{\alpha} (s_1 \xrightarrow{\gamma})^\omega$ ✓

d: Yes, infinite enabled and executed. ✓

e: Yes, same in (d)

f: Yes, same in (d)

F4:

a: Yes ~ 0.25

b: No, η is infinite enabled but not executed. ✓

c: No, $F4 = F2 \cap F3$ Even though F2 and F3 both satisfy (c), but in different executions. - 0.25

d: No, δ infinite enabled but not execute. ✓

e: Yes, execution $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\eta} (s_3 \xrightarrow{\eta})^\omega$ path? - 0.25

f: Yes, same in (e) ~ 0.25

Exercise 3: Fairness 4/6

1. weak fairness for "enter"

- Sufficiency. "Enter" is always enabled in "w", we don't want it to do self loop on "w", so "Enter" must infinite execute.
- Weakest. Weak Fairness is the weakest. Without weak fairness for "enter", it will do self loop on "w".

2. Strong fairness for "enter"

- Sufficiency. We don't want the case $(w_1 \rightarrow w_2 \rightarrow)^\omega$. If "Enter" is infinite enabled, it must infinite executed.
- Weakest. Weak fairness for "enter" is not possible. Because in the case $(w_1 \rightarrow w_2 \rightarrow)^\omega$ "Enter" is infinite enabled, but not always enabled.

3. Not possible to give fairness assumption on action enter to ensure non-starvation?

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